



NOPCOMMERCE OPENTELEMETRY

Instrumenting a real e-commerce platform end-to-end

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Software Architectures
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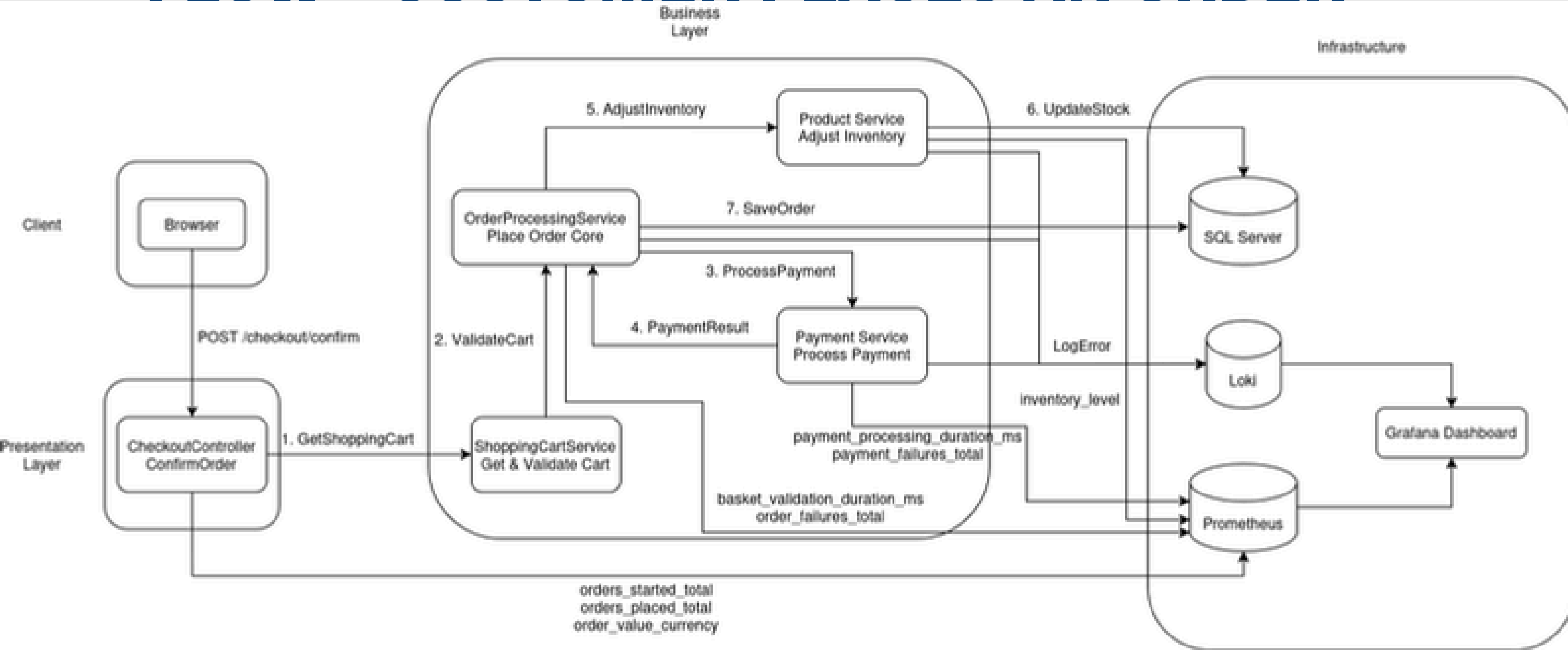
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FLOW - CUSTOMER PLACES AN ORDER



CUSTOM METRICS

inventory.level - Histogram

Stock after each

AdjustInventoryAsync, tagged by product.id. Surfaces stockout risk before customers see it.

orders.value - Histogram

Distribution of OrderTotal per order. Detects €0.00 pricing bugs and supports revenue SLOs — not just request counts.

order_failures_total - Counter

Covers all failure modes with flow_stage tag (payment / order_core / lock). A spike at lock = burst traffic; at payment = gateway degrading.

checkout_duration_ms - Histogram

End-to-end latency from confirm button to redirect. Rising P99 with stable P50 = tail latency from a downstream dep.

PRIVACY STRATEGY

Service Layer

Tags span with
customer.id,
order.total

DataProcessor

OnEnd(): replaces
[REDACTED]

Otel Collector

Receives only
clean spans

Jaeger/Loki

No PII ever
reaches storage

WHY AT EXPORT TIME?

- Single enforcement point — not scattered across 20+ tag sites
- Business logic untouched — no risk of breaking service code

REDACTED KEYS

email

password

credit

card

address

phone

ssn

payment

04 TRADEOFFS

Decision	Approach	Tradoff/Why
Redaction layer	SDK processor (export-time)	Centralised enforcement vs. per-tag discipline. Guarantees PII never leaves the process, but requires running the OTel SDK.
Span granularity	Boundary-only (entry/exit of service methods)	Minimal risk, low overhead. Loses visibility into internal stages, but those can be inferred from parent-child relationships and metrics.
Metrics type	Histograms for durations, counters for failures	Rich percentiles vs. simpler gauges. Adds cardinality (1e buckets) but enables SLOs and anomaly detection.
Propagation	OTLP/gRPC + Collector	Centralised control (filtering, sampling, routing) vs. direct export. Adds one more component to maintain.
Caching visibility	Not instrumented	No distinction between cache hit/miss in traces. Low priority because database spans already show query cost.

DEMO